

The population structure of invasive *Lantana camara* is shaped by its mating system

Reviewed Preprint

v1 • April 16, 2025

Not revised

P Praveen , Rajesh Gopal, Uma Ramakrishnan 

National Centre for Biological Sciences - TIFR, GKVK Campus, Bangalore, India • Global Tiger Forum – East of Kailash, New Delhi, India

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

eLife Assessment

The submission by Praveen and colleagues reports **important** findings describing the structure of genetic and colour variation in its native range for the globally invasive weed *Lantana camara*. Whilst the importance of the research question and the scale of the sampling is appreciated, the analysis, which is currently **incomplete**, requires further tests to support the claims made by the authors.

<https://doi.org/10.7554/eLife.104988.1.sa2>

Abstract

Over the last century, invasive species have emerged as an important driver of global biodiversity loss. Many invasive species have low genetic diversity in the invaded habitats, owing to the demographic bottleneck during introduction. *Lantana camara* is one of the hundred most problematic invasive species globally. Despite its ecological importance in many countries, our understanding of the genetic diversity patterns of this plant remains poor. Previous studies hypothesize that invasive *L. camara* is a species complex with a hybrid origin, but this remains untested. We investigated the population genetic patterns of this invasive species by sampling 359 plants that represented a spectrum of flower colour variants across 36 locations, spanning most of the biogeographic regions across India. Analyses of the population structure using 19,008 SNPs revealed that *L. camara* in India exhibits a strong genetic structure. Interestingly, the structuring pattern does not exhibit a strong correlation with geography. In the structure analysis, individuals with similar flower colours clustered together regardless of their location of origin. The genetic distance between most of the individuals was low, indicating the absence of multiple species. A high inbreeding coefficient and a low proportion of heterozygous sites observed suggested that the strong structure could be due to self-fertilization. Thus we infer that *L. camara* exists as homozygous inbred lines formed by self-fertilization and that these inbred lines could be associated with distinct flower colours. Together, this would explain the correlation between flower colour and genetic structure, and the lack of geographic structure. These results refute the argument that *L. camara* is a species complex and emphasize the importance of the mating system in shaping the patterns of diversity in this invasive species. Our findings highlight a hitherto

unknown role for mating systems in invasive species, furthering our understanding of evolution in invasive species.

1. Introduction

Invasive species are one of the major causes of ongoing biodiversity loss and extinctions in the Anthropocene (Linders et al., 2019; Mollot et al., 2017). Understanding the ecological and evolutionary underpinnings of invasion success is critical to controlling the emergence of new invasive species. Evolutionary and population genetic theory suggests that among other factors, genetic variation is critical to demographic success and range expansion (Ørsted et al., 2019). Populations that have undergone a demographic bottleneck are expected to have low genetic variation, which can negatively impact their ability to survive and adapt (Olazcuaga et al., 2023).

Invasive species typically undergo demographic bottlenecks during colonization (Alves et al., 2022). Low propagule pressure (the number of individuals introduced) leads to low effective population size. Consequently, invasive populations are established under conditions of limited genetic variation (Dlugosch & Parker, 2008; Simberloff, 2009). Despite the theoretical expectation of low genetic variation, invasive populations paradoxically demonstrate the ability to adapt, thrive and reproduce in large numbers in these novel habitats. This phenomenon is called the genetic paradox of biological invasions (Estoup et al., 2016). While extensively debated, some studies support the presence of such low genetic variation (Alves et al., 2022) while others contradict it (Bastien Lavergne & Molofsky, 2007; Zhang et al., 2021). The inherent complexities of human-mediated introduction like the choice of source population, number of introductions, artificial hybridizations etc. along with evolutionary forces make the patterns of genetic variation in invasive populations complicated (Roman & Darling, 2007). Several factors such as propagule pressure, hybridizations, gene flow, mating system etc. play an important role in determining the amount of genetic variation in invasive populations (Jaspers et al., 2021).

The introduction of many individuals either through single or multiple events can bring high variation into the novel environment (Bastien Lavergne & Molofsky, 2007). The source populations from which these individuals originate can influence the patterns of genetic diversity (Ryan et al., 2019) in new populations. Gene flow between multiple invasive populations can homogenize genetic variation (Vallejo-Marín et al., 2021). Intraspecific hybridizations between genetically distinct individuals can lead to the formation of new genotypes with novel allelic combinations, while hybridization with native species can aid in rapid adaptation (Blair & Hufbauer, 2010; Mesgarana et al., 2016). In plant populations, mating systems play a role in determining the pattern of genetic diversity (Hamrick et al., 1992; Ness et al., 2010; Schoen & Brown, 1991). Theoretically, cross-fertilizing species have high heterozygosity, and hence high genetic variation (Koelling et al., 2011). Conversely, with self-fertilization heterozygosity decreases by 50% every generation (Hedrick 2011). This leads to high genome-wide homozygosity and individuals exist as homozygous inbred lines (Guo et al., 2008; Pérez De La Vega & Garda, 1997). In self-fertilizing species, neutral diversity is low due to low effective population size (Foxe et al., 2009; Guo et al., 2008; Koelling et al., 2011), and it results in high genetic drift. The combination of low gene flow, high drift and high homozygosity leads to a strong genetic structure in self-fertilizing species compared to cross-fertilizing species (Edwards et al., 2019; Koelling et al., 2011; Mairal et al., 2023). While mating systems play an important role in determining genetic diversity, to the best of our knowledge, their role in invasive species has not been specifically investigated.

Here we present the genetic diversity patterns of the globally invasive plant species, *Lantana camara* (*Lantana* hereafter) in India. *Lantana* is a perennial shrub native to Central and South America. During the colonial era, this species was introduced into many countries across the

globe, like India, Australia and South Africa as a garden plant (Kannan et al., 2013). Currently, it is one of the most important invasive species in India with a significant impact on the native ecosystems. On average, Lantana threatens almost 44 per cent of the forest areas in India (Mungi et al., 2020). In the Bandipur tiger reserve in southern India, Lantana occupies at least 40 per cent of the land area (R et al., 2016). Unlike many invasive species, Lantana exhibits extensive phenotypic variation in India (Negi, Sharma, & Vishvakarma, 2019), with the most striking being the variation in flower colour. By 1900, approximately 16 species of the genus Lantana were reported in India, potentially suggesting multiple introductions. The taxonomy of the genus Lantana is complicated due to the presence of polyphyletic groups (Lu-Irving et al., 2021). In the taxonomy, many flower colour variants of Lantana are classified as distinct species (Sanders, 2006). However, genetic validation of this classification, focusing on different flower colour variants has not been done. It is possible that active hybridization to produce new hybrids took place in botanical gardens (Goyal & Sharma, 2015a; Urban et al., 2011). Previous studies have hypothesized that invasive Lantana is a species complex, potentially formed from such hybridization events (Goyal & Sharma, 2015b; Sanders, 2006). Variation in Lantana flower colour lends support to the species complex hypothesis. As a result, the invasive Lantana is sometimes referred to as *Lantana camara* sensu lato (Goyal & Sharma, 2015c). Detailed genetic studies are important in checking this hypothesis and characterising different species of this genus.

Despite its status as one of the most pervasive invasive species globally, our understanding of its biology and genetics remains limited. It is further complicated by uncertainties regarding its species identity. Investigating the genetic diversity patterns of Lantana presents an opportunity to improve our understanding regarding the uncertainties about the species identity, as well as the genetic paradox of biological invasions. This can help in improving our understanding of the taxonomy of Lantana itself. A previous population genetic study using microsatellite markers suggests that Lantana populations in India are genetically homogeneous (Ray & Quader, 2014). However, as discussed earlier, its ecological and demographic success is very high along with the high phenotypic variation. We use thousands of genome-wide Single Nucleotide Polymorphism (SNP) markers and genetic simulations to investigate the patterns of genetic diversity, spatial genetic structure and the factors shaping these patterns in Lantana populations in India. We use India as an example of a large, ecologically diverse landmass where Lantana has been highly successful. More specifically, we investigate i) the genetic diversity and population structure of Lantana populations in India, and ii) the evolutionary forces shaping the genetic diversity patterns of Lantana.

2. Material and methods

2.1 Study system

Lantana camara is considered to be one of the hundred most invasive species in the world (IUCN 2000). It was introduced across the globe as a garden plant because of its strikingly attractive flowers. The earliest recorded presence of Lantana in India comes from the British botanical garden in Kolkata in 1807 (Kannan et al., 2013). Currently, it is widely distributed in India occupying almost every terrestrial habitat (Mungi et al., 2020). In India, Lantana exhibits extensive phenotypic variation with the most remarkable being the variation in the flower colour (Negi, Sharma, Vishvakarma, et al., 2019). Lantana produces umbellate inflorescence with younger flowers in the centre and older ones in the periphery. The colour of the flowers may change post-pollination, thus a single inflorescence can have flowers of different colours (Negi, Sharma, & Vishvakarma, 2019). Yellow-pink (yellow young flowers in the centre of the inflorescence and older pink flowers in the periphery), White-pink and Orange (flowers maintain more or less the

same colour) flower colour types are commonly observed in India ([Fig. 1b](#)). Flower colours like reddish-pink, yellow-orange and pink ([Fig. 1b](#)) are relatively uncommon. A single location can have multiple of these flower colour variants growing together next to each other.

2.2 Sampling locations

Lantana plant samples were collected between the years 2020 and 2023. Plants were collected from 36 locations across the country, representing various bioclimatic regions ([Fig. 1a](#), Supplementary Table 1). Wild-growing plant saplings were collected and planted in the greenhouse at the National Center for Biological Sciences, Bengaluru. From each location, a few plants that represented different flower colour types were selected randomly for the genomic analysis. We received silica gel-dried leaf samples from four locations which were also included in the study. Whenever possible, we have noted the flower colour during collection, or in the greenhouse—however, the exact flower colour of 59 plants that died before flowering is unknown.

2.3 DNA extraction and sequencing library preparation

Two to three fresh leaves were collected from each plant for DNA extraction. Leaves were crushed using liquid nitrogen in a mortar and pestle. A Qiagen DNeasy plant pro kit was used for the extractions, following the manufacturer's instructions. DNA was quantified using a Qubit fluorometer. 199 μ l Qubit buffer and 1 μ l dye were mixed and 199 μ l of this mix was used with 1 μ l DNA to check the concentrations. Extracted DNA was stored at -20 degrees Celsius. Double-digest Restriction-site-associated DNA (ddRAD) libraries were prepared using the modified protocol by Tyagi et al 2024. Briefly, genomic DNA was digested using two restriction enzymes; SphI a six-base pair and MluCI a four-base pair cutter. Adapters specifically designed for SphI and MluCI are ligated to the cut ends. Unbound adapters were removed using magnetic bead-based cleaning, where 0.5X beads were used to remove fragments shorter than 50 base pairs. Individual specific index combinations were ligated to each sample using PCR and amplification was carried out. The PCR products were dual size selected for a target library size of 200 to 500 bp using magnetic beads. Library quality was evaluated using a TapeStation automated electrophoresis system. All the samples were pooled to get a 2nM library and the final library was sequenced in HiSeq2500 and NovaSeq6000 platforms.

2.4 Variant calling and filtering

Raw reads were demultiplexed and barcodes were trimmed using a custom script. Raw reads were quality-controlled using Trimmonatic - v.0.33.0 ([Bolger et al., 2014](#)). Reads with an average quality lower than 30 and reads shorter than 30 bp were excluded. Trimmed reads were aligned to a draft reference genome of *L. camara* ([Joshi et al, 2022](#)) using BWA-MEM ([Li & Durbin, 2009](#)) with default settings. We used Freebayes ([Garrison & Marth, 2012](#)) to call variants using the default settings, which were filtered using VCFtools ([Danecek et al., 2011](#)) to retain only high-quality SNPs. The filtering was done in the following order: biallelic SNPs were retained, SNPs with a minimum depth of 10 and a maximum depth of 500 were retained. Sites with a minimum genotyping quality of 30 and a minor allele count of at least three were kept. Individuals with more than 90% missing data were removed. SNPs that are not in Hardy-Weinberg equilibrium were removed. Finally, SNPs present in at least 90 per cent of the individuals were retained. The filtered VCF was used for all further analyses.

2.5 Population structure and genetic diversity estimations

We used a Bayesian clustering approach (ADMIXTURE) and a multivariate approach (Principal Component Analysis) to understand the genetic structuring of Lantana. Principal Component Analysis (PCA) was carried out using PLINK ([Purcell et al., 2007](#)). Genetic population structuring was investigated using the model-based clustering algorithm ADMIXTURE ([Alexander et al., 2009](#)), with K values ranging from 2 to 17. We used Evanno's delta K method ([Evanno et al.,](#)

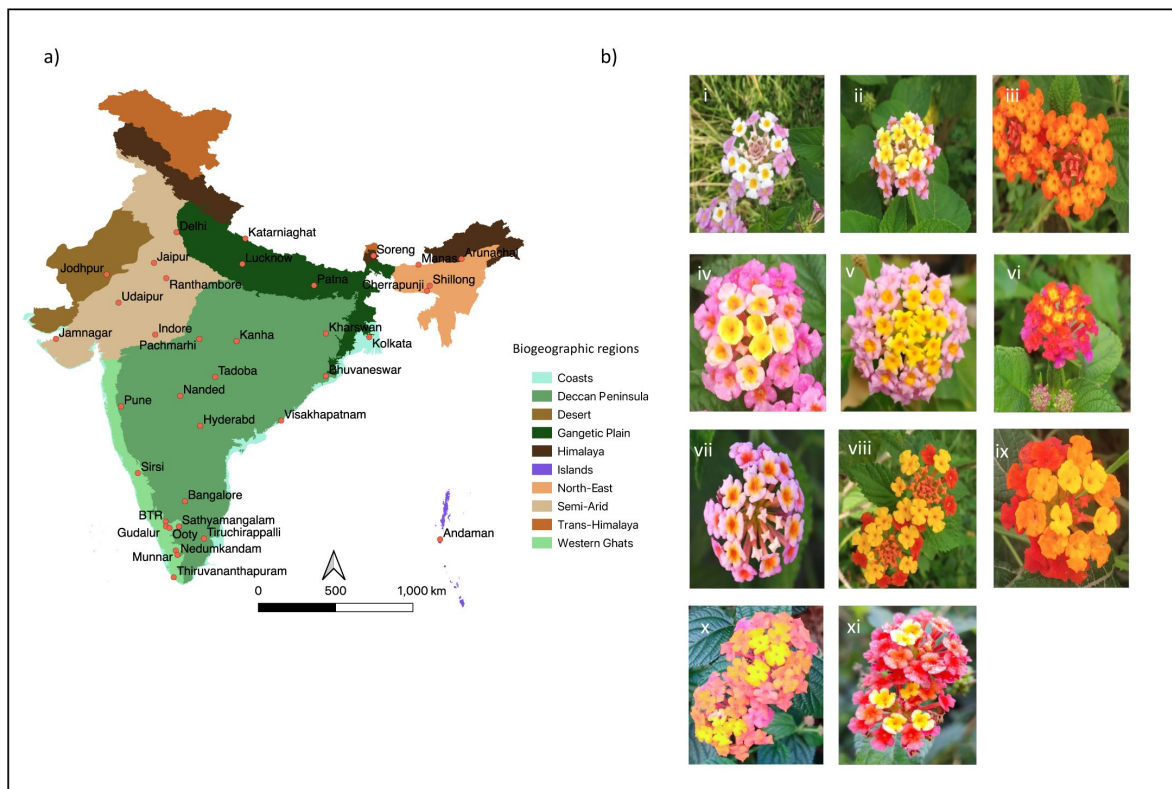


Figure 1

Sampling locations (a) of Lantana across different biogeographic regions in India Flower colour variation (b) of Lantana. i- White-pink, ii- Yellow-pink, iii- orange, iv- Yellow-pink-b, v - Yellow-pin-c, vi - Reddish-pink, vii- pink, viii- yellow-orange, ix - yellow-orange-b, x - Yellow-dark-pink a, xi - Yellow-dark-pink b

2005) implemented in the web server CLUMPAK (Kopelman et al., 2015) to find the optimum K. CLUMPAK was used to plot the admixture proportions as well. The structure plot was rearranged for easy visualization. Inbreeding coefficients and heterozygosity were estimated using the R package Adegenet (v2.1.10) (Jombart, 2008). F_{ST} was estimated using Weir and Cockerham's method implemented in VCFtools (Danecek et al., 2011). Adegenet (v2.1.10) was used to test for isolation by distance. Nei's genetic distance was calculated using the StAMPP package (Pembleton et al., 2013) in R. A phylogenetic network was generated using the R package StAMPP and the SplitsTree4 (Huson, 1997).

2.6 Genetic simulations

We used SLiM (Haller & Messer, 2019), a forward-in-time, individual-based simulation program to understand the importance of mating systems on the pattern of genetic diversity in invasive plant species. Populations analogous to both native and invasive populations were simulated with different mating systems. All the individuals in the simulations were diploid hermaphrodites, with 10 chromosomes each with 1,00,000 bps. The mutation rate and the recombination rate were assumed to be 10^{-8} and 10^{-7} respectively (Igor Kovalchuk, 2000), which is the general mutation and recombination rate for plants. Generations were assumed to be non-overlapping. Scenarios with different mating systems having varying levels of self-fertilization (zero - complete crossing, 25, 50, 75 -mixed-mating and 100-complete selfing) were simulated. A schematic representation of the demographic changes during the simulation is shown in **Figure 2**. The simulation started with a single population of 100 genetically identical individuals, which grew exponentially (growth rate=1.05) until they reached a carrying capacity of 10,000 individuals. This carrying capacity was decided, to reduce the computational complexity. In the burn-in phase, the individuals evolve under these conditions for 20,000 years/generations. In the 20,000th year, two new populations were created by individuals migrating from the initial simulated population (assumed to be the native population). The number of founding individuals (in this case propagule pressure) was either 10, 100 or 1000. These populations grew exponentially to a carrying capacity of 10,000 with the same growth rate. The population evolved till the year 20,200 and thus 200 generations after invasion is simulated. The population level heterozygosity, nucleotide diversity and F_{ST} between populations were measured at the end of the 20th and 200th generations. For randomly selected 50 individuals from each population, VCF files with SNP information were created at the 20,200th generation.

3 Results

3.1 Invasive Lantana populations in India show strong genetic structure

Using ddRAD sequencing, we obtained 19,008 SNP markers for 359 Lantana individuals from different regions across India. This SNP dataset was utilised to assess the genetic differentiation in Lantana through three different analyses: Structure, Principal component analyses and fixation index. Structure analyses using the ADMIXTURE package revealed strong genetic structure in the Indian Lantana populations (**Fig. 3a**). The optimum number of Hardy-Weinberg populations was between four and eleven (Supplementary Figure 1). At K=4, which had the highest support for the optimum number of clusters, the geographic origin did not dictate the genetic structuring. Despite belonging to the same geographic location, individuals were assigned across different genetic clusters. At K=3, nine individuals from North-Eastern India clustered together and maintained a distinctive identity at higher K as well. Many individuals were admixed. K=8 and 11 had the second and third highest log likelihood values for the optimum K.

Figure 2

Schematic of the SLiM simulations. In generation one, a single population is formed with 100 individuals. It exponentially grows and reaches a carrying capacity of 10,000. At generation 20,000 two new invasive populations are formed, founded by 10 individuals each. Further, all three populations evolve for 200 more years.

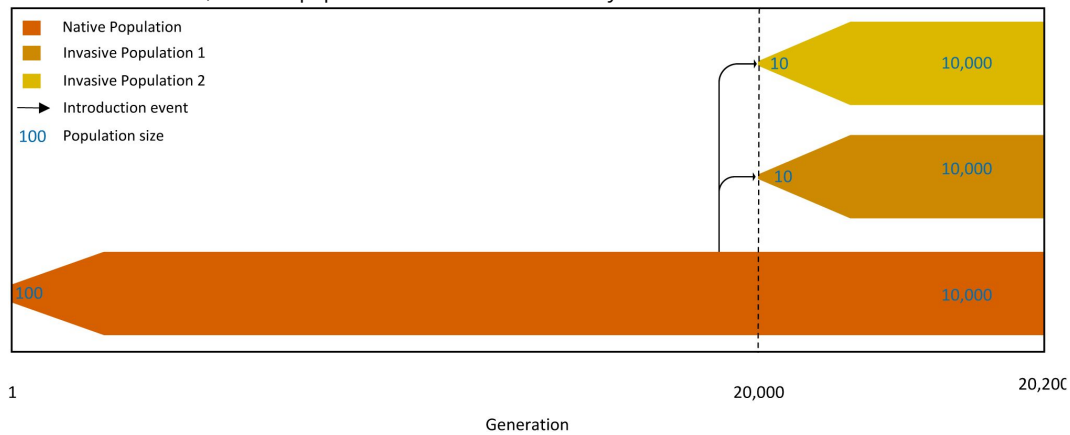
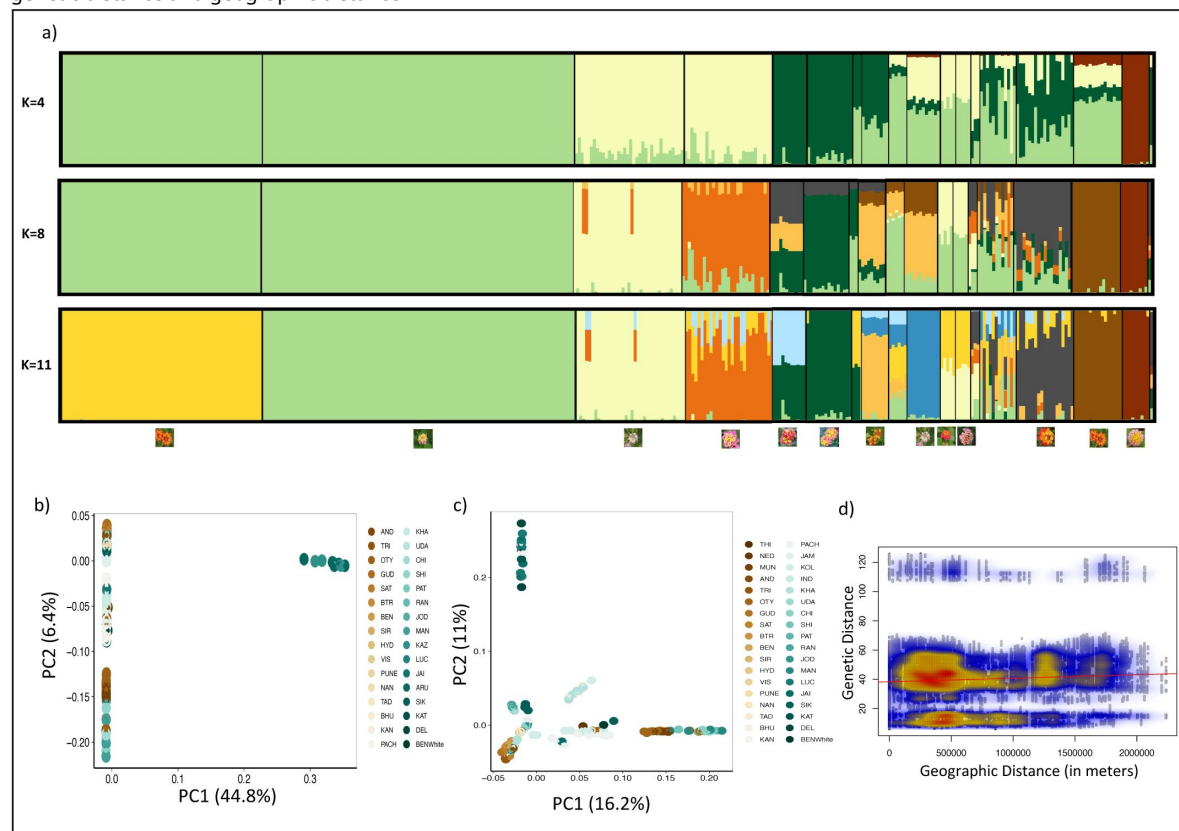


Figure 3

Population structure of Lantana in India. a) Genetic ancestry assignment based on the programme ADMIXTURE. Ancestry fractions assuming four, eight and eleven ancestral populations are given. Clusters are arranged based on the flower colour variants b) PCA of all 359 samples c) PCA of all samples except nine northeast Indian samples. Individuals are coloured based on their geographic locations from north to south of India d) Isolation by distance showing a weak correlation between genetic distance and geographic distance



The principal component analysis results corroborated the findings from the admixture analyses (Fig. 3b and c). In the PCA encompassing all 359 samples (Fig. 3b), PC 1 explained 44.8% of the variation and PC 2 explained 6.4% of the total variation. Nine samples from North-eastern India formed a distinct cluster separate from the rest of the samples in the PC1 axis (Fig. 3b). To explore the structuring pattern within the remaining samples, a separate analysis excluding these nine individuals was carried out (Fig. 3c). In this analysis, PC 1 explained 16.2% of the variation and PC 2 explained 11% of the variation. In the PCA, samples were coloured based on the geographic location from south to north of India (brown to green respectively). Individuals from geographically distant locations were clustering together while those from the same geographic area were assigned to different clusters in many locations, reinforcing the absence of a strong geography-based pattern. A flower colour-based clustering pattern was evident in the PCA with samples colour-coded according to the flower colour (Supplementary Figure 2).

The isolation by distance analysis using the Mantel test (Fig. 3d) revealed a weak correlation between genetic distance and geographic distance (Mantel $r = 0.039$, p -value = 0.075), indicating that geography is not the major factor influencing the structuring pattern. The fixation index (F_{ST}) between most of the population was low (Supplementary Table 2). Arunachal Pradesh (in north-eastern India) samples showed high F_{ST} with many populations (0.46 to 0.54) (Supplementary Table 2).

3.2 Genetic population structuring in *Lantana* is associated with flower colour

In the Admixture analysis, the clusters closely aligned with distinct flower colour variants of *Lantana*. At $K=4$, yellow-pink and some of the orange-flowered *Lantana* clustered together, while a separate cluster comprised pink and some of the white-pink individuals (Fig. 3a). A third cluster was formed by dark-pink-coloured plants from Ooty (Southern India) and North-east India. Nine individuals from northeast India formed a fourth cluster. Other flower colour variants were admixed. Interestingly, at $K=11$, a pronounced flower colour-based clustering pattern emerged, with distinct clusters formed by yellow-pink, pink, reddish-pink, dark pink and yellow-orange individuals, while both orange and white-pink *Lantana* each formed two separate clusters. In the PCA, the clustering pattern often mirrored the flower colour variants (Supplementary Figure 2). Yellow-pink colour variant formed a tight cluster. One of the orange colour variants also showed a similar trend. In the phylogenetic network generated using SplitsTree, (Fig. 4a) most of the individuals were grouped according to their flower colour. The most common flower colour in the country, yellow-pink, formed a single cluster in admixture analysis and in the SplitsTree network. Most of the yellow-pink and orange individuals across the country had short branch lengths in the phylogenetic network.

High Inbreeding and low genetic diversity characterize *Lantana* populations Nearly all the individuals exhibited a very low proportion of heterozygous sites, with less than 10% heterozygous sites (Fig. 4c). However, the nine individuals from North-East India showed the highest proportion of heterozygous sites (>0.3). The inbreeding coefficient of many of the individuals was extremely high, suggesting high levels of self-fertilization (Fig. 4d). The Nei's genetic distance between most of the *Lantana* individuals across the country was very low indicating the absence of genetically distinct lineages (Fig. 4b). Even with low genetic distance, there is often strong differentiation within a location. The genetic distance was high only for the nine individuals from the northeast of India with the rest of the samples. This was reflected in the phylogenetic network, where the nine northeast individuals formed a distinct group with long branch lengths (Fig. 4a).

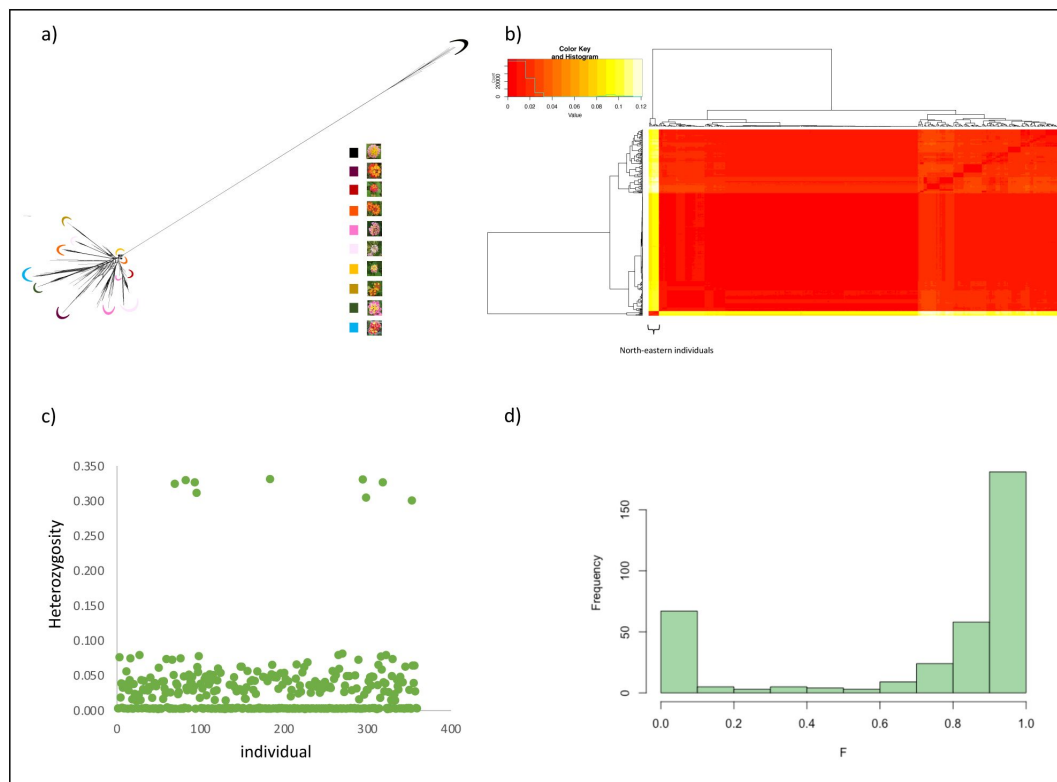


Figure 4

Signatures of high inbreeding due to self-fertilization in *Lantana* a) Phylogenetic network using SplitsTree4, lineages are coloured based on the flower colour b) Nei's genetic distance between all the individuals, nine northeast India samples are genetically most distinct c) Proportion of heterozygous sites per individual d) Inbreeding coefficient

3.3 Genetic diversity in invasive populations is influenced by the mating system

To understand the influence of the mating system on genetic diversity in invasive plant species, we have carried out an individual-based genetic simulation using SLiM. These simulations confirmed that the mating system strongly impacts standing genetic variation within a species. Simulations of cross-fertilisation resulted in higher heterozygosity and nucleotide diversity than those of self-fertilization (Fig. 5a and b). This was consistent across various propagule pressures (10, 100 and 1000) simulated. The heterozygosity and nucleotide diversity were low in the simulated invasive populations compared to the simulated native population (Fig. 5a and b). An increase in the propagule pressure did not substantially increase the genetic diversity. Under the same propagule pressure, simulations of cross-fertilization gave rise to invasive populations with high genetic diversity compared to scenarios with self-fertilization. The variance in heterozygosity and nucleotide diversity was high in the self-fertilizing simulation scenarios.

Mating systems had an impact on the differentiation of simulated native and invasive populations. F_{ST} between simulated native and invasive populations revealed that the F_{ST} value was lowest under complete cross-fertilization (Fig. 5c). The differentiation increased as the percentage of self-fertilization increased. Under simulations of complete self-fertilization, the variance was high and F_{ST} ranged from approximately 0.03 to 0.69. Whereas, in complete cross-fertilization, it ranged from 0.07 to 0.2. Similarly, in simulations of cross-fertilization, the differentiation between simulated invasive populations was low compared to self-fertilisation scenarios (Fig. 5d). The differentiation increased as the allowed selfing percentage increased. F_{ST} ranged from 0.05 to 0.98 in self-fertilizing simulation scenarios. Within each category of selfing percentage, an increase in the propagule pressure reduced the differentiation.

Self-fertilization leads to strong structuring within the simulated populations. PCA using SNPs from individuals of all three populations revealed a strong genetic structure in self-fertilization scenarios (Supplementary Figure 3). Whereas, in cross-fertilization scenarios, there was a weak structure (Supplementary Figure 3). Under selfing, individuals from the same population were assigned to distinct genetic clusters, and within a cluster, individuals from multiple populations were present.

4 Discussion

Biological invasions pose a serious global challenge, with millions of dollars spent annually on their control (IPBES, 2023). Yet, it is often difficult to control the emergence of new invasive species and their spread due to the lack of understanding of the invasion process and the factors contributing to invasion success. Critical to understanding invasion success is an understanding of standing genetic variation and population differentiation. We used genomic techniques to understand patterns of genetic diversity and differentiation in invasive *Lantana* populations in India. Our findings revealed that *Lantana* populations in India are genetically structured, but the structuring pattern was not strongly correlated with the geography. In the Admixture analysis, the genetic structure aligned with flower colour variation. Notably, many individuals exhibited a high inbreeding coefficient and a very low proportion of heterozygous sites. Genetic distance and phylogenetic network indicated that lineages are relatively similar. We suggest that the strong genetic structure despite the low genetic distance is because of the increased homozygosity due to self-fertilization. Through genetic simulations, we show that the mating system can impact genetic diversity and the structure of simulated, putatively invasive species.

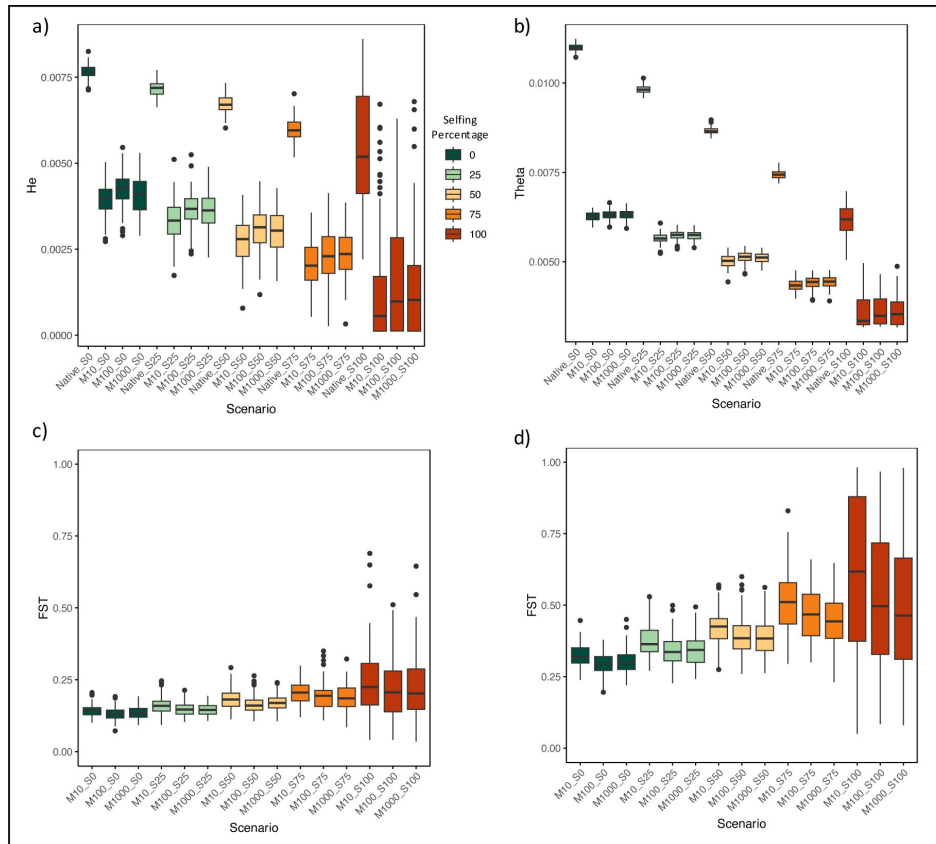


Figure 5

Individual-based genetic simulation results showing putative patterns of genetic diversity in invasive plants under different assumed mating systems and propagule pressure. Heterozygosity (a) and nucleotide diversity (b) under different mating systems in 'native' and 'invasive' populations founded by 10, 100 and 1000 individuals c) F_{ST} between 'native' and 'invasive populations' under simulated mating systems and propagule pressure. d) F_{ST} between two simulated 'invasive populations' under different mating systems and propagule pressure.

4.1 Population structure

Our investigation into the genetic structure of *Lantana* using both Admixture and principal component analysis revealed a strong population structure. Individuals from the same geographic location were assigned to different genetic clusters, contrary to a previous study using microsatellite markers which suggested genetic homogeneity in *Lantana* populations in India (Ray & Quader, 2014 [↗](#)). SNP markers are numerous in the genome, and as a result, have higher statistical power than microsatellite markers (Zimmerman et al., 2020 [↗](#)). We have used 19,008 SNP markers and thus we expected population structure to be better resolved.

In PCA, admixture, and phylogenetic network nine individuals from the northeast of India were genetically distinct compared to the rest of the samples. These samples exhibited high heterozygosity compared to other plants, which could explain their genetic distinctiveness. These plants also had noticeable phenotypic differences, such as a lack of thorns and dense trichomes on leaves. Further detailed analyses of these plants are required to better understand the basis of such a large genetic difference.

Many invasive species show strong genetic structure in the invaded ranges (Wang & Chen, 2017 [↗](#)) due to multiple introductions, followed by the lack of gene flow between populations (Bastien Lavergne & Molofsky, 2007 [↗](#)). However, in many such studies, a geography-based population structuring pattern is observed (Le Cam et al., 2020 [↗](#)). Our genetic results corroborate the expectation of multiple introductions as suggested by the historical records on *Lantana* (Kannan et al., 2013 [↗](#)). As expected with multiple introductions, we find multiple genetic clusters (across populations) and phenotypic (varied flower colours) diversity in invasive populations of *Lantana* in India. However, our findings are unique because, despite having multiple genetic clusters, we do not find geographic structuring.

4.2 Self-pollination leading to high homozygosity in *Lantana*

Most of the *Lantana* individuals in India show a high inbreeding coefficient and a low proportion of heterozygous sites. In plants, high inbreeding coefficient and low heterozygosity are associated with self-fertilization (Hedrick 2011). These results indicate that *Lantana* is predominantly self-fertilizing in India and exists as homozygous inbred lines. This is further supported by Structure and PCA results. In many natural populations of plants, the type of mating system is shown to influence the patterns of genetic diversity (Bomblies et al., 2010 [↗](#)). Mating systems in plants range from uniparental reproduction or self-fertilization to obligate cross-fertilization or outcrossing (Charlesworth, 2006 [↗](#)). Approximately 10-15% of seed plants predominantly self-fertilize (Goodwillie et al., 2005 [↗](#)). Many species exhibit mixed mating systems, including a combination of both selfing and outcrossing (Whitehead et al., 2018 [↗](#)). Mating systems influence the effective population size and thus play a crucial role in shaping genetic diversity within a species (Laenen et al., 2018 [↗](#)).

In its native range, *Lantana* primarily relies on cross-fertilization for reproduction (Barrows, 1976 [↗](#)). However, in the introduced range, *Lantana* is self-compatible (Goulson & Derwent, 2004 [↗](#); Ram & Mathur, 1984 [↗](#)). During colonization and invasion, plant species can shift mating systems from cross-fertilization to self-fertilization (Petanidou et al., 2012 [↗](#); Willi et al., 2022 [↗](#)). A detailed study of mating systems in native and invaded ranges is required to confirm such a shift in the case of *Lantana*. Self-fertilization is shown to increase the genome-wide homozygosity and inbreeding coefficient (Barragan et al., 2024 [↗](#)). The strong genetic structure even with low genetic differences can be attributed to the high homozygosity of the inbred lines formed through selfing (Bomblies et al., 2010 [↗](#)). The presence of multiple genetic groups in a single location is mainly due to the presence of such inbred lines. This is supported by studies like Bhattacharya et al., 2022 [↗](#). Selfing acts as a barrier to gene flow between these inbred lines, thereby maintaining their genetic identity. However, the genetic distance between these lines suggests that they are not genetically

very different. The presence of admixed individuals in many populations indicates the occurrence of cross-fertilization as well. These crossing events can help in generating new recombinant inbred lines. Future research including field-based studies on mating systems in *Lantana* can help better understand the extent and frequency of these various mating strategies.

4.3 Association of flower colour with genetic structure

Our study revealed an association between genetic clustering and flower colour variation. Different flower colour variants were assigned to distinct genetic clusters despite being from the same geographic location. This indicates the lack of gene flow between flower colour variants which grow physically close to each other in many locations. This correlation between flower colour and genetic structuring underscores the influence of the mating system. Here, self-fertilization acts as a barrier to gene flow and different inbred lines are associated with different flower colours. Thus, we think that the identity of these lines is maintained by self-fertilization. Historically, *Lantana* species were primarily defined mainly based on the flower colours and many such variants were considered as different species (Sanders, 2006 [↗](#)). Our study suggests that these are within species variation and the overall phenotypic similarity within a flower colour variant is maintained by self-fertilization. We think that the correlation between flower colour and genetic structure is caused by the formation of inbred lines. Further genetic analysis is required to understand the genetic basis of these flower colour variations.

4.4 Is *Lantana* a species complex?

The findings from the molecular analysis of *Lantana* challenge the notion of invasive *Lantana* being a species complex. While genetic structuring was evident, it predominantly arose from the prevalence of highly homozygous inbred lines rather than from the presence of genetically different species. Most lines exhibited low genetic distance between each other, indicating the high genetic similarity between flower colour variants. The high phenotypic variation observed could be the result of the introduction of multiple inbred lines to the subcontinent or the emergence of novel inbred lines through hybridization post-introduction. These variations could be within species phenotypic variations typical to many plant species. Through citizen science platforms like iNaturalist, it is evident that *Lantana* in its native range also exhibits similar flower colour variation. However, genomic analysis of related species of *Lantana* can help in understanding the chances of any past introgressions.

5 Conclusions

Studying genetic diversity patterns in invasive species is crucial for understanding their evolution. We explored genetic diversity patterns of the global invasive species *L. camara*. Our results reveal that *lantana* populations in India are strongly structured with genetic differences between flower colour variants. The low genetic distance observed between most plants negates the chances of *lantana* being a species complex. The presence of strong genetic structure even in the absence of high genetic distance is attributed to self-fertilization and the formation of putative inbred lines. This is supported by the low heterozygosity and high inbreeding coefficient. In predominantly self-fertilizing species, individuals exist as inbred lines, which restricts gene flow and allows these inbred lines to maintain their genetic identity. We suggest that the correlation between flower colour and genetic structure is due to the association of specific flower colours to specific inbred lines.

Uniquely, our study highlights the significance of the mating system in shaping the genetic diversity pattern in invasive plant species. Our results also highlight the importance of multiple introductions and the introduction of genetically different variants in contributing to high genetic

structuring within invasive populations. Further research on other invasive species will reveal how general these patterns might be.

Data availability statement

Accession numbers for the raw ddRAD sequencing data and the GitHub link for analysis codes will be provided upon acceptance of the manuscript.

Acknowledgements

We are thankful to Keval Palya, Mamta M., and Rajat Rastogi for their help with sampling and laboratory experiments. We are thankful to Divyashree Rana, Kritagnya Vadar, Nishma Dahal, Rachana Rao, Tikily Tayeng, Tista Ghosh and Vinay Sagar for aiding in the sample collection. We thank Mayuresh Gangal, Divyashree Rana, Abhinav Tyagi, Anubhab Khan, Vinay Sagar and Arpitha Jayanth for their valuable suggestions. We thank the forest departments of Karnataka (No: PCCF (WL) /E2/CR-52/2019-20), Tamil Nadu (No: WL(A)/52852/2019) and Madhya Pradesh (MP permit no:835, dated: 30-01-2020) for providing necessary permits. PP was supported by NCBS/TIFR (Department of Atomic Energy). This work was supported under DBT project No-BT/PR29251/FCB/125/18/2018. The NCBS data cluster used is supported under project 12-R&D-TFR-5.04-0900, Department of Atomic Energy, Government of India.

Additional information

Author contributions

Conceptualization: P.P., U. R, and R. G., Data collection: P.P., Laboratory work: P. P., Data analysis: P.P., Project administration: P.P., U.R., and R. G., Funding acquisition: U. R, and R. G., Writing original draft: P.P., Writing – Review and editing: U.R., Supervision: U.R.

Additional files

Supplementary information 

References

1. Joshi Adwait G., P. P. U. R. R. S. (2022) **Draft genome sequence of an invasive plant *Lantana camara* L** *Bioinformation* **18**:739–741 <https://doi.org/10.6026/97320630018739>
2. Alexander D. H., Novembre J., Lange K (2009) **Fast model-based estimation of ancestry in unrelated individuals** *Genome Research* **19**:1655–1664 <https://doi.org/10.1101/gr.094052.109>
3. Alves J. M., Carneiro M., Day J. P., Welch J. J., Duckworth J. A., Cox T. E., Letnic M., Strive T., Ferrand N., Jiggins F. M. (2022) **A single introduction of wild rabbits triggered the biological invasion of Australia** *Proc Natl Acad Sci U S A* **119**:e2122734119 <https://doi.org/10.1073/pnas.2122734119>
4. Barragan A. C., Collenberg M., Schwab R., Kersten S., Kerstens M. H. L., Požárová D., Bezrukov I., Bemm F., Kolár F., Weigel D (2024) **Deleterious phenotypes in wild *Arabidopsis arenosa* populations are common and linked to runs of homozygosity** *G3: Genes, Genomes, Genetics* **14** <https://doi.org/10.1093/g3journal/jkad290>
5. Barrows E. M. (1976) **Nectar Robbing and Pollination of *Lantana camara* (Verbenaceae)** *Biotropica* **8**:32–135
6. Bastien Lavergne S., Molofsky J. (2007) **Increased genetic variation and evolutionary potential drive the success of an invasive grass** <https://www.pnas.org/cgi/content/full/>
7. Bhattacharya S., Hernández F., Alves M. F., Machado R. M., Sun Y. Y., Wang M. R., Zhang C. Bin, Hao J. H. (2022) **Genetic diversity and population structure of invasive and native populations of *Erigeron canadensis* L** *Journal of Plant Ecology* **15**:864–876 <https://doi.org/10.1093/jpe/rtac016>
8. Blair A. C., Hufbauer R. A (2010) **Hybridization and invasion: One of North America's most devastating invasive plants shows evidence for a history of interspecific hybridization** *Evolutionary Applications* **3**:40–51 <https://doi.org/10.1111/j.1752-4571.2009.00097.x>
9. Bolger A. M., Lohse M., Usadel B (2014) **Trimmomatic: A flexible trimmer for Illumina sequence data** *Bioinformatics* **30**:2114–2120 <https://doi.org/10.1093/bioinformatics/btu170>
10. Bomblies K., Yant L., Laitinen R. A., Kim S. T., Hollister J. D., Warthmann N., Fitz J., Weigel D (2010) **Local-scale patterns of genetic variability, outcrossing, and spatial structure in natural stands of *Arabidopsis thaliana*** *PLoS Genetics* **6** <https://doi.org/10.1371/journal.pgen.1000890>
11. Charlesworth D (2006) **Evolution of Plant Breeding Systems** *Current Biology* **16** <https://doi.org/10.1016/j.cub.2006.07.068>
12. Danecek P., Auton A., Abecasis G., Albers C. A., Banks E., DePristo M. A., Handsaker R. E., Lunter G., Marth G. T., Sherry S. T., McVean G., Durbin R (2011) **The variant call format and VCFtools** *Bioinformatics* **27**:2156–2158 <https://doi.org/10.1093/bioinformatics/btr330>
13. Dlugosch K. M., Parker I. M (2008) **Founding events in species invasions: Genetic variation, adaptive evolution, and the role of multiple introductions** *Molecular Ecology* **17**:431–

449 <https://doi.org/10.1111/j.1365-294X.2007.03538.x>

14. Edwards C. E., Albrecht M. A., Bassüner B., Yatskievych G. A (2019) **Population genetic analysis reveals a predominantly selfing mating system and strong genetic structuring in a naturally fragmented, threatened plant** *Conservation Genetics* **20**:1437–1448 <https://doi.org/10.1007/s10592-019-01226-9>
15. Estoup A., Ravign V., Hufbauer R., Vitalis R., Gautier M., Facon B. (2016) **Is There a Genetic Paradox of Biological Invasion?** *Annual Review of Ecology, Evolution, and Systematics* **47**:51–72 <https://doi.org/10.1146/annurev-ecolsys-121415-032116>
16. Evanno G., Regnaut S., Goudet J (2005) **Detecting the number of clusters of individuals using the software STRUCTURE: A simulation study** *Molecular Ecology* **14**:2611–2620 <https://doi.org/10.1111/j.1365-294X.2005.02553.x>
17. Foxe J. P., Slotte T., Stahl E. A., Neuffer B., Hurka H., Wright S. I. (2009) **Recent speciation associated with the evolution of selfing in *Capsella*** *PNAS* **106**
18. Garrison E., Marth G (2012) **Haplotype-based variant detection from short-read sequencing** *arXiv* <http://arxiv.org/abs/1207.3907>
19. Goodwillie C., Kalisz S., Eckert C. G (2005) **The evolutionary enigma of mixed mating systems in plants: Occurrence, theoretical explanations, and empirical evidence** *Annual Review of Ecology, Evolution, and Systematics* **36**:47–79 <https://doi.org/10.1146/annurev.ecolsys.36.091704.175539>
20. Goulson D., Derwent L. C (2004) **Synergistic interactions between an exotic honeybee and an exotic weed: Pollination of *Lantana camara* in Australia** *Weed Research* **44**:195–202 <https://doi.org/10.1111/j.1365-3180.2004.00391.x>
21. Goyal N., Sharma G. P (2015a) ***Lantana camara* L. (sensu lato): An enigmatic complex** *NeoBiota* **25**:15–26 <https://doi.org/10.3897/neobiota.25.8205>
22. Goyal N., Sharma G. P. (2015b) ***Lantana camara* L. (sensu lato): an enigmatic complex** *NeoBiota* **25**:15–26 <https://doi.org/10.3897/neobiota.25.8205>
23. Goyal N., Sharma G. P. (2015c) ***Lantana camara* L. (sensu lato): An enigmatic complex** *NeoBiota* **25**:15–26 <https://doi.org/10.3897/neobiota.25.8205>
24. Guo Y.-L., Bechsgaard J. S., Slotte T., Neuffer B., Lascoux M., Weigel D., Schierup M. H. (2008) **Recent speciation of *Capsella rubella* from *Capsella grandiflora*, associated with loss of self-incompatibility and an extreme bottleneck** *Proc Natl Acad Sci U S A* **106**:5246–5251 <https://doi.org/10.1073/pnas.0808012106>
25. Haller B. C., Messer P. W (2019) **SLiM 3: Forward Genetic Simulations Beyond the Wright-Fisher Model** *Molecular Biology and Evolution* **36**:632–637 <https://doi.org/10.1093/molbev/msy228>
26. Hamrick J. L., Godt J. W., Sherman-Broyles S. L. (1992) **Factors influencing levels of genetic diversity in woody plant species** *New Forests* **6**
27. Huson D. H. (1997) **SplitsTree: analyzing and visualizing evolutionary data** *Bioinformatics* **14**:68–73 <https://doi.org/10.1093/bioinformatics/14.1.68>

28. IPBES. (2023) **The thematic assessment report on invasive alien species and their control** <https://doi.org/10.5281/zenodo.7430682>
29. Jaspers, et al. (2021) **contributed new reagents/analytic tools** **118** <https://doi.org/10.1073/pnas.2116211118>
30. Jombart T (2008) **Adegenet: A R package for the multivariate analysis of genetic markers** *Bioinformatics* **24**:1403–1405 <https://doi.org/10.1093/bioinformatics/btn129>
31. Kannan R., Shackleton C. M., Shaanker R. U (2013) **Reconstructing the history of introduction and spread of the invasive species, Lantana, at three spatial scales in India** *Biological Invasions* **15**:1287–1302 <https://doi.org/10.1007/s10530-012-0365-z>
32. Koelling V. A., Hamrick J. L., Mauricio R (2011) **Genetic diversity and structure in two species of Leavenworthia with self-incompatible and self-compatible populations** *Heredity* **106**:310–318 <https://doi.org/10.1038/hdy.2010.59>
33. Kopelman N. M., Mayzel J., Jakobsson M., Rosenberg N. A., Mayrose I (2015) **Clumpak: A program for identifying clustering modes and packaging population structure inferences across K** *Molecular Ecology Resources* **15**:1179–1191 <https://doi.org/10.1111/1755-0998.12387>
34. Laenen B., Tedder A., Nowak M. D., Toräng P., Wunder J., Wötzel S., Steige K. A., Kourmpetis Y., Odong T., Drouzas A. D., Bink M. C. A. M., Ågren J., Coupland G., Slotte T (2018) **Demography and mating system shape the genome-wide impact of purifying selection in Arabis alpina** *Proceedings of the National Academy of Sciences of the United States of America* **115**:816–821 <https://doi.org/10.1073/pnas.1707492115>
35. Le Cam S., Daguin-Thiébaud C., Bouchemousse S., Engelen A. H., Mieszkowska N., Viard F. (2020) **A genome-wide investigation of the worldwide invader Sargassum muticum shows high success albeit (almost) no genetic diversity** *Evolutionary Applications* **13**:500–514 <https://doi.org/10.1111/eva.12837>
36. Li H., Durbin R (2009) **Fast and accurate short read alignment with Burrows-Wheeler transform** *Bioinformatics* **25**:1754–1760 <https://doi.org/10.1093/bioinformatics/btp324>
37. Linders T. E. W., Schaffner U., Eschen R., Abebe A., Choge S. K., Nigatu L., Mbaabu P. R., Shiferaw H., Allan E (2019) **Direct and indirect effects of invasive species: Biodiversity loss is a major mechanism by which an invasive tree affects ecosystem functioning** *Journal of Ecology* **107**:2660–2672 <https://doi.org/10.1111/1365-2745.13268>
38. Lu-Irving P., Bedoya A. M., Salimena F. R. G., dos Santos Silva T. R., Viccini L. F., Bitencourt C., Thode V. A., Cardoso P. H., O’Leary N., Olmstead R. G. (2021) **Phylogeny of Lantana, Lippia, and related genera (Lantaneae: Verbenaceae)** *American Journal of Botany* **108**:1354–1373 <https://doi.org/10.1002/ajb2.1708>
39. Mairal M., García-Verdugo C., Le Roux J. J., Chau J. H., van Vuuren B. J., Hui C., Münzbergová Z., Chown S. L., Shaw J. D. (2023) **Multiple introductions, polyploidy and mixed reproductive strategies are linked to genetic diversity and structure in the most widespread invasive plant across Southern Ocean archipelagos** *Molecular Ecology* **32**:756–771 <https://doi.org/10.1111/mec.16809>
40. Mesgarana M. B., Lewis M. A., Adesc P. K., Donohued K., Ohadia S., Lia C., Cousensa R. D (2016) **Hybridization can facilitate species invasions, even without enhancing local**

adaptation *Proceedings of the National Academy of Sciences of the United States of America* **113**:10210–10214 <https://doi.org/10.1073/pnas.1605626113>

41. Mollot G., Pantel J. H., Romanuk T. N. (2017) **The Effects of Invasive Species on the Decline in Species Richness: A Global Meta-Analysis** In: *Advances in Ecological Research* Academic Press Inc pp. 61–83
42. Mungi N. A., Qureshi Q., Jhala Y. V (2020) **Expanding niche and degrading forests: Key to the successful global invasion of *Lantana camara* (sensu lato)** *Global Ecology and Conservation* **23** <https://doi.org/10.1016/j.gecco.2020.e01080>
43. Negi G. C. S., Sharma S., Vishvakarma S. C. R (2019) **Ecology and Use of *Lantana camara* in India** *Bot. Rev.* **85**:109–130 <https://doi.org/10.1007/s12229-019-09209-8>
44. Negi Sharma, Vishvakarma S., Samant S. S. (2019) **Ecology and Use of *Lantana camara* in India** *Botanical Review* **85**:109–130 <https://doi.org/10.1007/s12229-019-09209-8>
45. Ness R. W., Wright S. I., Barrett S. C. H (2010) **Mating-system variation, demographic history and patterns of nucleotide diversity in the tristylous plant *Eichhornia paniculata*** *Genetics* **184**:381–392 <https://doi.org/10.1534/genetics.109.110130>
46. Olazcuaga L., Lincke B., DeLacey S., Durkee L. F., Melbourne B. A., Hufbauer R. A (2023) **Population demographic history and evolutionary rescue: Influence of a bottleneck event** *Evolutionary Applications* **16**:1483–1495 <https://doi.org/10.1111/eva.13581>
47. Ørsted M., Hoffmann A. A., Sverrisdóttir E., Nielsen K. L., Kristensen T. N (2019) **Genomic variation predicts adaptive evolutionary responses better than population bottleneck history** *PLoS Genetics* **15** <https://doi.org/10.1371/journal.pgen.1008205>
48. Pembleton L. W., Cogan N. O. I., Forster J. W (2013) **StAMPP: An R package for calculation of genetic differentiation and structure of mixed-ploidy level populations** *Molecular Ecology Resources* **13**:946–952 <https://doi.org/10.1111/1755-0998.12129>
49. Pérez De La Vega M., Garda P. (1997) **Genetic structure of self-pollinating species: the case of wild *Avena*** *Bocconeia*
50. Petanidou T., Godfree R. C., Song D. S., Kantsa A., Dupont Y. L., Waser N. M (2012) **Self-compatibility and plant invasiveness: Comparing species in native and invasive ranges** *Perspectives in Plant Ecology, Evolution and Systematics* **14**:3–12 <https://doi.org/10.1016/j.ppees.2011.08.003>
51. Purcell S., Neale B., Todd-Brown K., Thomas L., Ferreira M. A. R., Bender D., Maller J., Sklar P., De Bakker P. I. W., Daly M. J., Sham P. C. (2007) **PLINK: A tool set for whole-genome association and population-based linkage analyses** *American Journal of Human Genetics* **81**:559–575 <https://doi.org/10.1086/519795>
52. Vardhan V., Madan R., Kannan S., Arunkumar B., Prakash G. (2016) **Mapping the Distribution of *Lantana Camara* in the Bandipur Tiger Reserve, Karnataka** Draft Report submitted to Karnataka Forest Department March 2016
53. Ram H. Y. M., Mathur G (1984) **Flower Colour Changes in *Lantana camara*** *Journal of Experimental Botany* **35**:1656–1662

54. Ray A., Quader S (2014) **Genetic diversity and population structure of *Lantana camara* in India indicates multiple introductions and gene flow** *Plant Biol J* **16**:651–658 <https://doi.org/10.1111/plb.12087>
55. Roman J., Darling J. A. (2007) **Paradox lost: genetic diversity and the success of aquatic invasions** *Trends in Ecology and Evolution* **22**:454–464 <https://doi.org/10.1016/j.tree.2007.07.002>
56. Ryan S. F., Lombaert E., Espeset A., Vila R., Talavera G., Dinca V., Doellman M. M., Renshaw M. A., Eng M. W., Hornett E. A., Li Y., Pfrender M. E., Shoemaker D. W (2019) **Global invasion history of the agricultural pest butterfly *Pieris rapae* revealed with genomics and citizen science** *Proceedings of the National Academy of Sciences of the United States of America* **116**:20015–20024 <https://doi.org/10.1073/pnas.1907492116>
57. Sanders R. W (2006) **Taxonomy of *Lantana* sect. *Lantana* (Verbenaceae): I. Correct application of *Lantana camara* and associated names** *Sida* **22**
58. Schoen D. J., Brown A. H. D. (1991) **Intraspecific variation in population gene diversity and effective population size correlates with the mating system in plants (inbreeding species/genetic conservation/neutral model/maximium likelihood/isozyme)** *Proc. Natl. Acad. Sci. USA* **88** <https://www.pnas.org>
59. Simberloff D (2009) **The role of propagule pressure in biological invasions** *Annual Review of Ecology, Evolution, and Systematics* **40**:81–102 <https://doi.org/10.1146/annurev.ecolsys.110308.120304>
60. Urban A. J., Simelane D. O., Retief E., Heystek F., Williams H. E., Madire L. G (2011) **The Invasive ‘*Lantana camara* L.’ Hybrid Complex (Verbenaceae): A Review of Research Into Its Identity and Biological Control in South Africa** *African Entomology* **19**:315–348 <https://doi.org/10.4001/003.019.0225>
61. Vallejo-Marín M., Friedman J., Twyford A. D., Lepais O., Ickert-Bond S. M., Streisfeld M. A., Yant L., van Kleunen M., Rotter M. C., Puzey J. R. (2021) **Population genomic and historical analysis suggests a global invasion by bridgehead processes in *Mimulus guttatus*** *Communications Biology* **4** <https://doi.org/10.1038/s42003-021-01795-x>
62. Wang G., Chen Z (2017) **Strong population genetic structure of an invasive species, *Rhynchophorus ferrugineus* (Olivier), in southern China** *Ecol Evol.* **7**:10770–10781 <https://doi.org/10.1002/ece3.3599>
63. Whitehead M. R., Lanfear R., Mitchell R. J., Karron J. D. (2018) **Plant mating systems often vary widely among populations** *Frontiers in Ecology and Evolution* **6** <https://doi.org/10.3389/fevo.2018.00038>
64. Willi Y., Lucek K., Bachmann O., Walden N (2022) **Recent speciation associated with range expansion and a shift to self-fertilization in North American *Arabidopsis*** *Nature Communications* **13** <https://doi.org/10.1038/s41467-022-35368-1>
65. Zhang L., Hua Y., Wei S (2021) **High genetic diversity of an invasive alien species: Comparison between fur-farmed and feral american mink (neovison vison) in China** *Animals* **11**:1–11 <https://doi.org/10.3390/ani11020472>
66. Zimmerman S. J., Aldridge C. L., Oyler-McCance S. J (2020) **An empirical comparison of population genetic analyses using microsatellite and SNP data for a species of conservation concern** *BMC Genomics* **21** <https://doi.org/10.1186/s12864-020-06783-9>

Author information

P Praveen

National Centre for Biological Sciences - TIFR, GKVK Campus, Bangalore, India

ORCID iD: [0009-0008-4737-0638](https://orcid.org/0009-0008-4737-0638)

For correspondence: praveenprakash@ncbs.res.in

Rajesh Gopal

Global Tiger Forum – East of Kailash, New Delhi, India

Uma Ramakrishnan

National Centre for Biological Sciences - TIFR, GKVK Campus, Bangalore, India

ORCID iD: [0000-0002-5370-5966](https://orcid.org/0000-0002-5370-5966)

For correspondence: uramakri@ncbs.res.in

Editors

Reviewing Editor

Alexandre Fournier-Level

The University of Melbourne, Parkville, Australia

Senior Editor

Detlef Weigel

Max Planck Institute for Biology Tübingen, Tübingen, Germany

Reviewer #1 (Public review):

Summary:

The authors investigated the population structure of the invasive weed *Lantana camara* from 36 localities in India using 19,008 genome-wide SNPs obtained through ddRAD sequencing.

Strengths:

The manuscript is well-written, the analyses are sound, and the figures are of great quality.

Weaknesses:

The narrative almost completely ignores the fact that this plant is popular in horticultural trade and the different color morphs that form genetic populations are most likely the result of artificial selection by humans for certain colors for trade, and not the result of natural selfing. Although it may be possible that the genetic clustering of color morphs is maintained in the wild through selfing, there is no evidence in this study to support that. The high levels of homozygosity are more likely explained as a result of artificial selection in horticulture and relatively recent introductions in India. Therefore, the claim of the title that "the population structure.. is shaped by its mating system" is in part moot, because any population structure is in large part shaped by the mating system of the organism, but further misleading because it is much more likely artificial selection that caused the patterns observed.

<https://doi.org/10.7554/eLife.104988.1.sa1>

Reviewer #2 (Public review):**Summary:**

The authors performed a series of population genetic analyses in *Lantana camara* using 19,008 genome-wide SNPs data from 359 individuals in India. They found a clear population structure that did not show a geographical pattern, and that flower color was rather associated with population structure. Excess of homozygosity indicates a high selfing rate, which may lead to fixation of alleles in local populations and explain the presence of population structure without a clear geographic pattern. The authors also performed a forward simulation analysis, theoretically confirming that selfing promotes fixation of alleles (higher F_{st}) and reduction in genetic diversity (lower heterozygosity).

Strengths:

Biological invasion is a critical driver of biodiversity loss, and it is important to understand how invasive species adapt to novel environments despite limited genetic diversity (genetic paradox of biological invasion). *Lantana camara* is one of the hundred most invasive species in the world (IUCN 2000), and the authors collected 359 plants from a wide geographical range in India, where *L. camara* has invaded. The scale of the dataset and the importance of the target species are the strengths of the present study.

Weaknesses:

One of the most critical weaknesses of this study would be that the output modelling analysis is largely qualitative, which cannot be directly comparable to the empirical data. The main findings of the SLiM-based simulation were that selfing promotes the fixation of alleles and the reduction of genetic diversity. These are theoretically well-reported knowledge, and such findings themselves are not novel, although it may have become interesting these findings are quantitatively integrated with their empirical findings in the studied species. In that sense, a coalescent-based analysis such as an Approximate Bayesian Computation method (e.g. DIY-ABC) utilizing their SNPs data would be more interesting. For example, by ABC-based methods, authors can infer the split time between subpopulations identified in this study. If such split time is older than the recorded invasion date, the result supports the scenario that multiple introductions may have contributed to the population structure of this species. In the current form of the manuscript, multiple introductions were implicated but not formally tested.

I also have several concerns regarding the authors' population genetic analyses. First, the authors removed SNPs that were not in Hardy-Weinberg equilibrium (HWE), but the studied populations would not satisfy the assumption of HWE, i.e., random mating, because of a high level of inbreeding. Thus, the first screening of the SNPs would be biased strongly, which may have led to spurious outputs in a series of downstream analyses. Second, in the genetic simulation, it is not clear how a set of parameters such as mutation rate, recombination rate, and growth rate were determined and how they are appropriate. Importantly, while authors assume the selfing rate in the simulation, selfing can also strongly influence the effective mutation rate (e.g. Nordborg & Donnelly 1997 Genetics, Nordborg 2000 Genetics). It is not clear how this effect is incorporated in the simulation. Third, while the authors argue the association between flower color and population structure, their statistical associations were not formally tested. Also, it is not mentioned how flower color polymorphisms are defined. Could it be possible to distinguish many flower color morphs shown in Figure 1b objectively? I am concerned particularly because the authors also mentioned that flower color may change temporally and that a single inflorescence can have flowers of different colors (L160).

<https://doi.org/10.7554/eLife.104988.1.sa0>

Author response:

We sincerely thank the editor and both reviewers for their time and thoughtful feedback on our manuscript. We have addressed several of the concerns in the responses below and are currently working on additional analyses to further strengthen the study. These results will be incorporated into the final version of the research paper.

Reviewer #1 (Public review):

Summary:

*The authors investigated the population structure of the invasive weed *Lantana camara* from 36 localities in India using 19,008 genome-wide SNPs obtained through ddRAD sequencing.*

Strengths:

The manuscript is well-written, the analyses are sound, and the figures are of great quality.

Weaknesses:

The narrative almost completely ignores the fact that this plant is popular in horticultural trade and the different color morphs that form genetic populations are most likely the result of artificial selection by humans for certain colors for trade, and not the result of natural selfing. Although it may be possible that the genetic clustering of color morphs is maintained in the wild through selfing, there is no evidence in this study to support that. The high levels of homozygosity are more likely explained as a result of artificial selection in horticulture and relatively recent introductions in India. Therefore, the claim of the title that "the population structure.. is shaped by its mating system" is in part moot, because any population structure is in large part shaped by the mating system of the organism, but further misleading because it is much more likely artificial selection that caused the patterns observed.

The reviewer raises the possibility that the observed genetic patterns may have originated through the selection of different varieties by the horticultural industry. While it is plausible that artificial selection can lead to the formation of distinct morphs, the presence of a strong structure between them in the wild populations cannot be explained just based on selection. In the wild, different flower colour variants frequently occur in close physical proximity and should, in principle, allow for cross-fertilization. Over time, this gene flow would be expected to erode any genetic structure shaped solely by past selection. However, our results show no evidence of such a breakdown in structure. Despite co-occurring in immediate proximity, the flower colour variants maintain distinct genetic identities. This suggests the presence of a barrier to gene flow, likely maintained by the species' mating system. Moreover, the presence of many of these flower colour morphs in the native range—as documented through observations on platforms like iNaturalist—suggests that these variants may have a natural origin rather than being solely products of horticultural selection.

While it is plausible that horticultural breeding involved efforts to generate new varieties through crossing—resulting in the emergence of some of the observed morphs—even if this were the case, the dynamics of a self-fertilizing species would still lead to rapid genetic structuring. Following hybridization, just a few generations of selfing are sufficient to produce inbred lines, which can then maintain distinct genetic identities. As discussed in our manuscript, such inbred lines could be associated with specific flower colour morphs and persist through predominant self-fertilization. This mechanism provides a compelling explanation for the strong genetic structure observed among co-occurring flower colour variants in the wild.

While a recent bottleneck may have increased inbreeding, the strong and consistent genetic structuring we observe within populations is more indicative of predominant self-fertilization. To further validate this, we conducted a bagging experiment on *Lantana camara* inflorescences to exclude insect-mediated cross-pollination. The results showed no significant difference in seed set between bagged and open-pollinated flowers, supporting the conclusion that *L. camara* is primarily self-fertilizing in India.

As the reviewer rightly points out, the mating system of a species plays a crucial role in shaping patterns of genetic structure. However, in many natural populations, structuring patterns are often influenced by a combination of factors such as selection, barriers to gene flow, and genetic drift. In some cases, the mating system exerts a more prominent influence at the microgeographic level, while in others, it can shape genetic structure at broader spatial scales. What is particularly interesting in our study is that - the mating system appears to shape genetic structure at a subcontinental scale. Despite the species having undergone other evolutionary forces—such as a genetic bottleneck and expansion due to its invasive nature—the mating system exerts a more pronounced effect on the observed genetic patterns, and the influence of the mating system is remarkably strong, resulting in a clear and consistent genetic structure across populations.

Reviewer #2 (Public review):

Summary:

*The authors performed a series of population genetic analyses in *Lantana camara* using 19,008 genome-wide SNPs data from 359 individuals in India. They found a clear population structure that did not show a geographical pattern, and that flower color was rather associated with population structure. Excess of homozygosity indicates a high selfing rate, which may lead to fixation of alleles in local populations and explain the presence of population structure without a clear geographic pattern. The authors also performed a forward simulation analysis, theoretically confirming that selfing promotes fixation of alleles (higher F_{st}) and reduction in genetic diversity (lower heterozygosity).*

Strengths:

*Biological invasion is a critical driver of biodiversity loss, and it is important to understand how invasive species adapt to novel environments despite limited genetic diversity (genetic paradox of biological invasion). *Lantana camara* is one of the hundred most invasive species in the world (IUCN 2000), and the authors collected 359 plants from a wide geographical range in India, where *L. camara* has invaded. The scale of the dataset and the importance of the target species are the strengths of the present study.*

Weaknesses:

One of the most critical weaknesses of this study would be that the output modelling analysis is largely qualitative, which cannot be directly comparable to the empirical data. The main findings of the SLiM-based simulation were that selfing promotes the fixation of alleles and the reduction of genetic diversity. These are theoretically well-reported knowledge, and such findings themselves are not novel, although it may have become interesting these findings are quantitatively integrated with their empirical findings in the studied species. In that sense, a coalescent-based analysis such as an Approximate Bayesian Computation method (e.g. DIY-ABC) utilizing their SNPs data would be more interesting. For example, by ABC-based methods, authors can infer the split time between subpopulations identified in this study. If such split time is older than the recorded invasion date, the result supports the scenario that multiple introductions may have contributed to the population structure of this species. In the current form of the manuscript, multiple introductions were implicated but not formally tested.

Through our SLiM simulations, we aimed to demonstrate that a pattern of strong genetic structure within a location—similar to what we observed in *Lantana camara*—can arise under a predominantly self-fertilizing mating system. These simulations were not parameterized using species-specific data from *Lantana* but were intended as a conceptual demonstration of the plausibility of such patterns under selfing using SNP data. While the theoretical consequences of self-fertilisation have been widely discussed, relatively few studies have directly modelled these patterns using SNP data. Our SLiM simulations contribute to this gap and support the notion that the observed genetic structuring in *Lantana* may indeed result from predominant self-fertilisation.

We thank the reviewer for the suggestion regarding the use of simulations based on genomic data from *Lantana* and for explaining the importance of it. We are currently conducting demographic simulations using genomic data from *Lantana* to estimate divergence times between the different flower colour variants. We believe this analysis will offer deeper insights and provide further clarity on the points raised by the reviewers.

I also have several concerns regarding the authors' population genetic analyses. First, the authors removed SNPs that were not in Hardy-Weinberg equilibrium (HWE), but the studied populations would not satisfy the assumption of HWE, i.e., random mating, because of a high level of inbreeding. Thus, the first screening of the SNPs would be biased strongly, which may have led to spurious outputs in a series of downstream analyses.

Hardy-Weinberg Equilibrium (HWE) filtering is a commonly used step in SNP filtering analysis to exclude loci potentially under selection, thereby enriching for neutral variants and minimizing bias in downstream analyses. To ensure that our results are not influenced by selection-driven SNPs, we conducted the analysis both with and without applying the HWE filter. Notably, the number of SNPs retained did not drop significantly after filtering, and the overall patterns observed remained consistent across both approaches.

Second, in the genetic simulation, it is not clear how a set of parameters such as mutation rate, recombination rate, and growth rate were determined and how they are appropriate. Importantly, while authors assume the selfing rate in the simulation, selfing can also strongly influence the effective mutation rate (e.g. Nordborg & Donnelly 1997 Genetics, Nordborg 2000 Genetics). It is not clear how this effect is incorporated in the simulation.

The aim of the SLiM simulation was to demonstrate that the extreme genetic structuring observed in *Lantana camara* can plausibly arise in natural systems under predominant self-fertilization. For the simulation, we used mutation and recombination rates estimated for *Arabidopsis thaliana*, as these parameters are currently unknown for *Lantana*. The details of this will be added in the revised version, and thanks to the reviewer for pointing this out. While we acknowledge that this simulation does not provide an exact representation of the species' evolutionary history, the goal of the simulation was not to produce precise estimates but rather to illustrate the feasibility of such strong genetic structuring resulting from self-fertilization alone. The impact of the selfing on the mutation rate is not incorporated in the simulations now. We will look into the details of this.

Third, while the authors argue the association between flower color and population structure, their statistical associations were not formally tested.

We recognize that one of the key improvements needed for the manuscript is to provide experimental evidence supporting self-fertilization. To address this, we conducted a bagging experiment on *Lantana camara* inflorescences to prevent insect visitation and eliminate

insect-mediated cross-fertilization. The results showed no significant difference in seed set between bagged and open-pollinated inflorescences, indicating that Lantana is predominantly self-fertilizing in India. This finding is consistent with our genetic data and will be included in the revised version of the manuscript.

Also, it is not mentioned how flower color polymorphisms are defined. Could it be possible to distinguish many flower color morphs shown in Figure 1b objectively? I am concerned particularly because the authors also mentioned that flower color may change temporally and that a single inflorescence can have flowers of different colors (L160).

The different flower colour variants are visually distinguishable. Our classification of these variants is not based on the colour of individual flowers at a single time point, but rather on the overall colour change pattern across the inflorescence over time. In other words, the temporal aspect of colour change has been considered in our grouping. For example, in the “yellow-pink” variant, flowers begin as yellow when young and gradually turn pink as they age. Importantly, variants that follow this pattern do not transition to an orange type at any stage, which distinguishes them from other colour types. The varieties that don't change colours are named based on the single flower colour like “orange”.

<https://doi.org/10.7554/eLife.104988.1.sa3>